

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250285551

Kind Code

A1

Publication Date

September 11, 2025

Inventor(s)

Guedes; Jose Carlos Oliveira

SYSTEM AND METHOD FOR STIMULATING THE ACTIVATION OF NEURAL MIRRORING MECHANISMS USING PERSONALIZED AVATARS FOR EDUCATION, LEARNING, PROFESSIONAL TRAINING, AND THERAPEUTIC APPLICATIONS

Abstract

A system and method for stimulating the activation of neural mirroring mechanisms through personalized avatars to enhance learning and skill development. The system creates hyper-realistic or stylized digital representations of users and animates these avatars to present educational content or demonstrate skills, stimulating mirror neuron activation when users observe themselves demonstrating mastery. Applications include academic learning, language acquisition, professional training, personal development, athletic performance enhancement, and therapeutic interventions including for Autism Spectrum Disorder.

Inventors: Guedes; Jose Carlos Oliveira (CLERMONT, FL)

Applicant: Guedes; Jose Carlos Oliveira (CLERMONT, FL)

Family ID: 96949294

Appl. No.: 19/220399

Filed: May 28, 2025

Publication Classification

Int. Cl.: G09B9/00 (20060101); **G06T13/40** (20110101); **G09B19/04** (20060101); **G10L13/047** (20130101)

U.S. Cl.:

CPC G09B9/00 (20130101); **G06T13/40** (20130101); **G09B19/04** (20130101); **G10L13/047** (20130101);

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims the benefit of U.S. Provisional Application No. 63/795,733, Receipt Date: Apr. 18, 2025, the entire contents of which are hereby incorporated by reference.

FIELD OF THE INVENTION

[0002] The present invention relates generally to educational technology and therapeutic interventions, and more specifically to a system and method for enhancing learning outcomes and therapeutic efficacy through the stimulation of natural neural mirroring mechanisms using hyper-realistic or stylized personalized avatars and voice cloning technology across educational, professional, and therapeutic domains.

BACKGROUND OF THE INVENTION

[0003] Traditional approaches to education and therapy face significant challenges in engagement, personalization, and effectiveness across multiple domains including academic learning, professional training, language acquisition, and therapeutic interventions. Despite technological advances, many digital learning platforms and therapeutic tools still rely on conventional approaches transferred to digital formats, without leveraging the unique capabilities of technology to enhance the neurological processes involved in learning and skill development.

[0004] The concept of neural mirroring, also known as mirror neurons or mirroring mechanisms, has been well-documented in neuroscience literature. Mirror neurons were first discovered in the early 1990s by researchers at the University of Parma, Italy, led by Giacomo Rizzolatti (Rizzolatti et al., 1996), who observed that certain neurons in macaque monkeys fired both when the monkey performed an action and when it observed another individual performing the same action.

Subsequent research by Gallese et al. (2004) and Iacoboni et al. (2005) has identified similar mirroring mechanisms in humans, suggesting that these neural systems play a crucial role in learning, imitation, understanding others' actions and intentions, and developing social cognition.

[0005] Research by Oberman et al. (2007) and Ramachandran & Oberman (2006) has demonstrated that neural mirroring mechanisms are fundamental to various learning processes, from motor skill acquisition to language learning, academic knowledge retention, and social skill development. These mechanisms allow humans to learn efficiently through observation and imitation, creating neural pathways that facilitate skill acquisition and knowledge retention. Studies by Rizzolatti & Craighero (2004) have shown that the stronger the activation of mirror neuron systems during learning, the more effective the learning process becomes.

[0006] Current educational technologies have attempted to leverage various cognitive and neurological principles to enhance learning, including spaced repetition, active recall, and multimedia learning. However, as noted by Meltzoff et al. (2009), few have specifically targeted the neural mirroring mechanism as a central component of the learning process. Existing platforms that utilize avatars or virtual representations typically present generic characters or instructors rather than personalized representations of the learners themselves, limiting their potential to activate mirror neuron systems effectively.

[0007] In language learning applications, educational platforms such as Duolingo, Babbel, and Rosetta Stone often rely on repetition, gamification, and external motivation factors, but do not effectively engage the neural mirroring mechanisms that could potentially accelerate learning and improve retention. Additionally, while some platforms offer personalization in terms of content and pacing, they do not personalize the learning experience at the neurological level by allowing learners to observe themselves as the source of knowledge.

[0008] In professional training contexts, current approaches to skill development often fail to create the deep neural connections that facilitate rapid skill acquisition and retention. Traditional training methods, even when digitized through platforms like LinkedIn Learning or Coursera, typically do not leverage the power of self-observation and neural mirroring to accelerate learning and improve performance.

[0009] In therapeutic contexts, including but not limited to interventions for Autism Spectrum Disorder (ASD), virtual reality and avatar-based systems have begun to emerge as tools for social skills training. However, these systems typically use pre-designed avatars or characters rather than personalized representations of the users themselves. This limits their effectiveness in activating mirror neuron systems, as research by Uddin et al. (2007) suggests that self-recognition enhances neural mirroring processes.

[0010] Significantly, research by Dapretto et al. (2006) and Williams et al. (2006) has demonstrated that individuals with Autism Spectrum Disorder often exhibit deficits in mirror neuron functionality. Studies using electroencephalography (EEG), functional magnetic resonance imaging (fMRI), and other neuroimaging techniques have shown reduced activation in mirror neuron regions when individuals with ASD observe others' actions compared to neurotypical individuals. This "broken mirror" theory of autism suggests that impaired mirror neuron function may underlie many of the social and communicative difficulties characteristic of ASD, including challenges with imitation, understanding others' intentions, empathy, and social reciprocity.

[0011] Despite the proliferation of educational and therapeutic technologies, a comprehensive review of existing platforms by Hsu (2017) found that none have specifically designed their systems to maximize the activation of neural mirroring mechanisms through the use of personalized avatars of the users themselves. While some platforms like Bitmoji or Memoji allow users to create cartoon-like representations of themselves, these are not used in a systematic way to stimulate neural mirroring for educational or therapeutic purposes.

[0012] It would not be obvious to a person skilled in the art to utilize self-based hyper-realistic or stylized avatars for the purpose of neurologically activating mirror neurons, as prior art focuses on gamification or content customization without engaging the user's own neurobiological mirroring pathways. While avatar technology and educational platforms exist separately, their combination specifically to stimulate the user's neural mirroring mechanisms represents a non-obvious technical innovation that produces unexpected results in learning acceleration and retention. The prior art fails to recognize the neurological significance of self-observation in the learning process, instead focusing on external representations and generic avatars that do not effectively engage the mirror neuron system's natural response to self-recognition.

[0013] Despite the widespread availability of both avatar technology and educational platforms, the specific combination of hyper-realistic or stylized self-avatars with educational content delivery represents a non-obvious technical solution that addresses the fundamental neurological mechanisms of learning in ways not contemplated by existing systems. The technical problem of effectively engaging mirror neuron systems during digital learning has remained unsolved, with conventional approaches failing to recognize the critical distinction between observing generic instructors versus observing oneself demonstrating mastery.

[0014] There remains a significant need for a comprehensive technology platform that specifically leverages and stimulates the neural mirroring mechanism through personalized avatars to enhance outcomes across multiple domains: academic learning, language acquisition, professional training,

personal development, athletic performance enhancement, and various therapeutic interventions including but not limited to those for individuals with Autism Spectrum Disorder.

SUMMARY OF THE INVENTION

[0015] The present invention provides a system and method for stimulating the activation of natural neural mirroring mechanisms to enhance learning and skill development across multiple domains using personalized avatars. The invention creates a unique learning experience by allowing users to observe their own realistic or stylized avatar teaching content or demonstrating skills, thereby stimulating the activation of neural mirroring mechanisms that facilitate accelerated learning, improved retention, and enhanced skill development in academic, professional, personal, athletic, and therapeutic contexts.

[0016] In one aspect, the invention comprises a computer-implemented system that captures the user's likeness through photos or videos, creates a realistic or stylized avatar of the user, and then animates this avatar to present educational content or demonstrate skills. The system synchronizes the avatar's lip movements, facial expressions, and gestures with the content, creating the impression that the user is watching themselves demonstrate mastery of the subject matter or skills being learned.

[0017] In another aspect, the invention provides a method for enhancing learning and skill development by stimulating the activation of neural mirroring mechanisms through self-observation. By seeing their own avatar demonstrate mastery of the subject matter or skills, users experience enhanced neural activation patterns that facilitate faster learning, better retention, and improved performance compared to traditional methods.

[0018] The system includes modules for avatar creation, voice cloning, content management, facilitation of natural neural mirroring, progress tracking, and adaptive learning. It can be applied to various domains, including:

[0019] Academic Learning: The system supports comprehensive education in various subjects including mathematics, science, history, literature, and more, with the user's avatar presenting complex concepts in an engaging and personalized manner;

[0020] Language Learning: The system provides immersive language acquisition for multiple languages, with initial focus on English, Spanish, and Portuguese, allowing users to see and hear their own avatar speaking fluently in the target language;

[0021] Professional Training: The system facilitates skill development in business communication, leadership, technical skills, and other professional competencies, with the user's avatar demonstrating expert performance;

[0022] Personal Development: The system supports acquisition of life skills, habit formation, and personal growth, with the user's avatar modeling desired behaviors and mindsets;

[0023] Athletic Performance: The system enhances sports training and physical skill development by showing the user's avatar demonstrating proper techniques, strategies, and mental preparation methods; and

[0024] Therapeutic Applications: The system includes specialized protocols for various therapeutic needs, including but not limited to social skill development for individuals with Autism Spectrum Disorder, speech therapy, cognitive rehabilitation, and treatment of anxiety disorders.

[0025] For individuals with Autism Spectrum Disorder, the system offers a novel approach to addressing mirror neuron deficits, providing a tool that can be used in conjunction with traditional therapies to enhance outcomes. The system's ability to create a safe, controlled environment for practicing social interactions through one's own avatar may help reduce anxiety and increase engagement, two common challenges in ASD interventions.

[0026] A key innovation of the invention is its implementation of a content exchange and distribution framework that allows educational and therapeutic institutions to provide specialized content through the platform, creating a B2B relationship indirectly through B2C interactions. This allows schools, universities, language centers, therapy providers, and individual teachers to upload their own educational or therapeutic content, which users can access using their personalized avatars.

[0027] The invention provides significant advantages over existing educational technologies and

interventions by specifically targeting and stimulating the activation of neural mirroring mechanisms through personalized avatar-based learning, resulting in improved engagement, accelerated learning, enhanced retention of content, and improved skill development across multiple domains.

[0028] Unlike existing platforms that may use generic avatars or pre-recorded instructors, this invention is the first to systematically use hyper-realistic or stylized representations of the users themselves specifically to stimulate neural mirroring mechanisms. This approach creates a stronger neurological connection and more effective learning pathway than is possible with traditional methods or existing technologies.

[0029] The invention's adaptive learning system ensures that content difficulty and presentation are continuously adjusted based on user performance, maintaining an optimal challenge level that maximizes learning outcomes while minimizing frustration.

[0030] The system architecture is designed to incorporate generative artificial intelligence models trained on user data to create adaptive content presentations. These models analyze user performance patterns, learning preferences, and engagement metrics to dynamically generate personalized educational content that is then presented through the user's avatar. The integration of generative AI capabilities enhances the system's ability to provide precisely tailored learning experiences while maintaining the neural mirroring benefits of self-observation.

[0031] Overall, the present invention represents a significant advancement in educational technology and skill development by stimulating the activation of neural mirroring mechanisms through personalized avatars to enhance learning, retention, and performance across academic, professional, personal, athletic, and therapeutic domains.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0032] The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate embodiments of the invention and, together with the description, serve to explain the principles of the invention.

[0033] FIG. 1: is a block diagram illustrating the overall system architecture of the TwinsDo platform, showing the primary components and their interactions.

[0034] FIG. 2: is a flowchart depicting the avatar creation process, from initial image capture to animation-ready personalized avatar.

[0035] FIG. 3: is a diagram illustrating the neural mirroring mechanism, showing how the user's personalized avatar stimulates mirror neuron pathways during the learning process.

[0036] FIG. 4: is a flowchart showing the therapeutic protocol for users with Autism Spectrum Disorder, illustrating the progression from basic to advanced social skill development.

[0037] FIG. 5: is a user interface mockup showing the main dashboard of the TwinsDo system, with sections for educational content, therapeutic activities, progress tracking, and marketplace access.

[0038] FIG. 6: is a diagram illustrating the content delivery process, showing how educational or therapeutic content is transformed and presented through the user's personalized avatar.

[0039] FIG. 7: is a flowchart depicting the adaptive learning system, showing how user performance data influences content selection and difficulty adjustment.

[0040] FIG. 8: is a diagram showing the B2B via B2C marketplace model, illustrating how educational and therapeutic institutions interact with individual users through the TwinsDo platform.

[0041] FIG. 9: is a series of screenshots demonstrating a language learning session, showing the user's avatar teaching vocabulary, pronunciation, and grammar.

[0042] FIG. 10: is a series of screenshots demonstrating a therapeutic session for emotion recognition, showing the user's avatar displaying various emotional expressions with corresponding labels and descriptions.

[0043] FIG. 11: is a diagram illustrating the data collection and privacy protection mechanisms implemented in the TwinsDo system.

[0044] FIG. 12: is a comparison chart showing learning outcomes between traditional methods and the TwinsDo neural mirroring approach, based on preliminary research data.

DETAILED DESCRIPTION OF THE INVENTION

Overview

[0045] Clarification of Neural Mirroring Mechanism: For the purposes of this invention, the term “neural mirroring mechanism” refers to a naturally occurring neurocognitive process, in which certain brain regions, commonly associated with mirror neurons, are activated when an individual observes actions associated with themselves or others. This mechanism is not implemented in software or hardware, but is instead triggered naturally in the brain through audiovisual conditions provided by the invention. The system facilitates the activation of this effect through hyper-realistic digital self-representation and synchronized personalized voice animation.

[0046] The TwinsDo system provides a novel approach to education, skill development, and therapeutic intervention by stimulating the activation of neural mirroring mechanisms through personalized avatars. The system creates realistic or stylized digital representations of users and animates these avatars to present educational content or demonstrate skills, creating the impression that users are watching themselves teach the material they are trying to learn or demonstrate the skills they are trying to develop.

[0047] This approach is based on established neuroscientific principles regarding mirror neurons and neural mirroring mechanisms, which have been shown to play a significant role in learning, particularly in observational learning and imitation. By observing their own avatar demonstrating mastery of the subject matter or skills, users experience enhanced neural activation patterns that facilitate faster learning, better retention, and improved performance.

[0048] The Neural Mirroring mechanism refers to a scientifically documented natural neurological process, in which mirror neurons in the brain are activated when an individual observes an action being performed, especially when that action is performed by a representation of oneself. This system is not technically implemented in the software, but rather naturally activated in the user's brain when certain conditions are created by the platform.

System Architecture

[0049] The TwinsDo system comprises several integrated components: [0050] Avatar Creation Module: Captures and processes user images to create realistic or stylized digital avatars that resemble the user; [0051] Voice Cloning System: Captures and processes user voice samples to create a digital voice model that simulates or approximates the user's own voice; [0052] Content Management System: Organizes educational and therapeutic content by subject, difficulty level, and learning or therapeutic objectives; [0053] Avatar Animation System: Animates the user's avatar to present educational content or demonstrate skills with appropriate lip synchronization, facial expressions, and gestures; [0054] Therapeutic Protocol System: Provides specialized content and progression paths for various therapeutic applications, including but not limited to social skill development for individuals with ASD; [0055] Progress Tracking System: Monitors user performance, learning outcomes, and progress across different domains. The system incorporates generative artificial intelligence models trained on user data to create adaptive content presentations. These models analyze user performance, learning patterns, and preferences to dynamically generate personalized educational content that optimizes engagement and knowledge retention; [0056] Adaptive Learning Algorithm: Adjusts content difficulty and presentation based on user progress and specific needs; [0057] Content Exchange and Distribution Framework: Enables educational and therapeutic institutions to provide and sell specialized content through the

system; and [0058] Adaptive Content Generation Module: Utilizes generative artificial intelligence to create personalized educational content based on user performance and preferences.

[0059] The system is implemented as a web-based application using cloud infrastructure, with client-side components for user interaction and server-side components for processing, storage, and content delivery.

[0060] System Integration Flow: Upon user registration and onboarding, the TwinsDo platform captures biometric data (image and voice) to generate a hyper-realistic or stylized avatar with a personalized digital voice model. When a learning or therapeutic session is initiated, the content management system retrieves relevant modules based on the user's progress and objectives. The adaptive engine selects appropriate content difficulty and pacing. Simultaneously, the avatar animation system synchronizes visual gestures, lip movement, and expressions with the generated speech. The session is delivered through the user interface in a format where the user observes their own avatar demonstrating mastery of skills or knowledge. Throughout the session, engagement metrics are collected by the progress tracking module and fed back into the adaptive system to refine future sessions. All components operate within a secure cloud-based infrastructure, ensuring scalability, real-time rendering, and privacy-compliant data handling.

Public API and SDK Integration

[0061] To extend the reach and applicability of the TwinsDo system, the platform includes a Public Application Programming Interface (API) and Software Development Kit (SDK). These components allow third-party developers, educational institutions, therapy providers, and training platforms to integrate their content and services directly into the TwinsDo environment.

[0062] Through the public API, external systems can: [0063] Push customized educational or therapeutic content to the platform; [0064] Retrieve anonymized or user-consented learning and therapeutic progress data; [0065] Trigger avatar-based playback sessions with externally provided scripts or exercises; and [0066] Integrate TwinsDo's adaptive learning engine into third-party learning environments.

[0067] The SDK provides pre-configured tools and libraries that enable the development of modules, activities, or experiences that operate within the TwinsDo avatar framework. This modular architecture encourages an ecosystem of external contributors while preserving the neurological benefits of the avatar-based mirror neuron stimulation.

Avatar Creation Process

[0068] The avatar creation process involves the following steps: [0069] Image Capture: The user provides photographs or video of themselves through a webcam or smartphone camera. Multiple angles and expressions are captured to create a comprehensive model; [0070] Image Processing: The system processes these images using computer vision algorithms to create a 3D model of the user's face and head, identifying key facial features and characteristics; [0071] Avatar Generation: Using digital imaging techniques and neural network-based algorithms, the system creates a realistic or stylized avatar that resembles the user while being suitable for animation; [0072] Customization: The user can make adjustments to their avatar's appearance, including clothing, hairstyle, and background settings, to enhance comfort and engagement; and [0073] Animation Preparation: The system prepares the avatar for animation by mapping facial features and creating control points for lip synchronization, expressions, and gestures.

[0074] For users with sensory sensitivities or preferences, the avatar creation process includes additional customization options to adjust visual and auditory elements to comfortable levels.

Avatar Licensing and External Use Export

[0075] The TwinsDo platform includes functionality for avatar export and licensing, enabling users to securely use their hyper-realistic or stylized avatars outside the platform in permitted environments. This includes integration with third-party video tools, social platforms, conferencing systems, and personal content creation workflows.

[0076] To ensure privacy and brand integrity, all exported avatar content is cryptographically

signed and watermarked. Licenses may include limitations based on domain of use, duration, and revocability, and may be managed through smart contracts or usage tokens issued by the TwinsDo platform.

[0077] This functionality allows the user's self-avatar to maintain consistency and continuity across various platforms, while preserving the underlying therapeutic and pedagogical objectives of the system.

Voice Cloning Process

[0078] The voice cloning process involves the following steps: [0079] Voice Sample Collection: The user provides voice samples through microphone recordings. Multiple samples with varying intonations and speech patterns are captured to create a comprehensive voice model; [0080] Audio Processing: The system processes these recordings using audio analysis algorithms to identify key voice characteristics, including pitch, tone, rhythm, and unique speech patterns; [0081] Voice Model Generation: Using deep learning techniques, particularly recurrent neural networks (RNNs) or transformer models, the system creates a voice model that can synthesize speech that simulates or approximates the user's voice; [0082] Text-to-Speech Integration: The voice model is integrated with text-to-speech functionality, allowing any text to be converted into speech that sounds like the user's voice; and [0083] Refinement and Customization: The user can provide feedback on the voice model, and the system can make adjustments to improve accuracy and naturalness.

[0084] For users with speech disorders or accents, the voice cloning process can include options to normalize certain speech characteristics while maintaining the unique identifiable qualities of the user's voice.

Neural Mirroring Stimulation

[0085] The core innovation of the TwinsDo system is its ability to stimulate the activation of natural neural mirroring through personalized avatars. This is achieved through: [0086] Personalized Representation: By using the user's own likeness, the system creates a stronger neural connection than would be possible with generic avatars, potentially enhancing mirror neuron activation as demonstrated by research from Uddin et al. (2007); [0087] Voice Personalization: By using the user's own cloned voice, the system creates an auditory connection that enhances the self-recognition effect and strengthens neural mirroring; [0088] Realistic Animation: The avatar's movements, particularly lip synchronization and facial expressions, are designed to be realistic and natural, enhancing the self-recognition effect; [0089] First-Person Perspective: The content is presented as if the user's avatar is teaching directly to the user, creating a self-reflective learning experience that maximizes neural mirroring; and [0090] Multimodal Presentation: The system combines visual, auditory, and interactive elements to engage multiple neural pathways simultaneously, enhancing learning and retention.

[0091] The neural mirroring engine synchronizes the avatar's lip movements with the educational or therapeutic content, ensuring that the avatar appears to be speaking naturally. Facial expressions and gestures are added to enhance engagement and convey emotional context appropriate to the learning material or skill being demonstrated.

Security and Privacy Protection

[0092] The system implements biometric security protocols to ensure that each avatar is uniquely bound to its rightful user, preventing unauthorized reproduction of personal likeness. These protocols include: [0093] Cryptographic binding of biometric identifiers to user accounts; [0094] Watermarking technology embedded within avatar rendering to prevent unauthorized duplication; [0095] Secure enclave storage of biometric templates with multi-factor authentication requirements for access; and [0096] Continuous verification protocols that monitor for unauthorized avatar usage or reproduction attempts.

Educational Applications

[0097] The TwinsDo system supports a wide range of educational applications, leveraging the stimulation of neural mirroring mechanisms to enhance learning outcomes: [0098] Language

Learning: The system provides comprehensive language learning for multiple languages, with the user's avatar demonstrating proper pronunciation, grammar, and conversational skills; [0099] Academic Subjects: Content covers various academic disciplines including mathematics, science, history, and literature, presented through the user's avatar to enhance engagement and retention; [0100] Professional Skills: The system includes modules for business communication, public speaking, leadership, and other professional competencies; and [0101] Exam Preparation: Specialized content helps users prepare for standardized tests and professional certifications, with the avatar demonstrating problem-solving strategies and test-taking techniques.

[0102] For language learning applications, the system includes: [0103] Vocabulary acquisition through contextual learning; [0104] Pronunciation practice with visual feedback; [0105] Grammar instruction with interactive exercises; [0106] Conversational practice in simulated scenarios; and [0107] Cultural context and idiomatic expressions.

[0108] For academic subjects, the system structures content according to established educational standards and curricula, allowing seamless integration with formal educational programs.

[0109] The exam preparation modules are specifically designed to address test anxiety by having the user's avatar demonstrate confidence and competence, potentially activating mirror neuron pathways associated with these positive emotional states.

Therapeutic Applications

[0110] The TwinsDo system includes specialized therapeutic applications designed for various needs: [0111] Social Skills Development: The system provides a progressive curriculum of social skills, from basic facial expression recognition to complex social interactions, presented through the user's own avatar; [0112] Emotion Recognition Training: Users observe their own avatar displaying various emotions, helping to strengthen neural pathways associated with emotion recognition and empathy; [0113] Speech Therapy: The system demonstrates proper articulation and speech patterns, with the user's avatar modeling correct pronunciation and speech techniques; [0114] Cognitive Rehabilitation: For individuals with cognitive deficits, the system provides exercises for attention, memory, executive functions, and problem-solving skills; and [0115] Anxiety and Phobia Treatment: Through gradual exposure therapy, users can observe their avatar successfully navigating anxiety-provoking situations.

[0116] For individuals with Autism Spectrum Disorder, the system includes specialized protocols that address the mirror neuron deficits often associated with ASD: [0117] Conversation Modeling: The system demonstrates appropriate conversational exchanges, with the user's avatar modeling both sides of the interaction to enhance understanding of social reciprocity; [0118] Perspective Taking: Through specialized exercises where the user's avatar interacts with others, the system helps develop theory of mind and perspective-taking abilities; and [0119] Behavioral Rehearsal: Users can practice social scenarios in a safe, controlled environment before encountering them in real life, reducing anxiety and building confidence.

[0120] The therapeutic protocols are designed to be customizable based on the individual's specific needs, abilities, and progress. Content difficulty and complexity are automatically adjusted based on performance, ensuring an optimal challenge level that promotes learning without causing frustration.

[0121] Progress tracking for therapeutic applications includes detailed metrics on: [0122] Engagement time and frequency; [0123] Performance on recognition and interaction tasks; [0124] Progress through skill development sequences; [0125] Generalization of skills across different scenarios; and [0126] Areas of strength and challenge

[0127] These metrics can be shared with therapists, educators, and caregivers (with appropriate permissions) to complement traditional interventions and inform treatment planning.

Professional Training Applications

[0128] The TwinsDo system offers comprehensive professional training applications: [0129] Communication Skills: The system helps users develop effective verbal and written communication

skills, with their avatar demonstrating techniques for clarity, persuasion, and engagement; [0130] Leadership Development: Users can observe their avatar demonstrating leadership behaviors, decision-making processes, and team management techniques; [0131] Sales and Customer Service: The system provides training in customer interaction, sales techniques, and conflict resolution, with the user's avatar modeling successful approaches; and [0132] Technical Skills: For specific industries, the system can demonstrate technical procedures, software usage, and specialized workflows.

[0133] The professional training modules are designed to accelerate skill acquisition through the stimulation of neural mirroring mechanisms, allowing users to internalize complex skills more rapidly than through traditional training methods.

Athletic Performance Applications

[0134] The TwinsDo system provides specialized applications for enhancing athletic performance:

[0135] Technique Demonstration: The user's avatar demonstrates proper form, technique, and execution for various sports and physical activities; [0136] Strategic Training: The system presents tactical and strategic concepts through the user's avatar, enhancing understanding and decision-making in competitive contexts; [0137] Mental Preparation: The avatar guides users through visualization exercises, pre-competition routines, and mental resilience techniques; and [0138] Performance Analysis: The system can incorporate actual performance data to show the user's avatar executing techniques with corrected form and improved efficiency.

[0139] The athletic performance applications leverage the neural mirroring mechanism to accelerate skill acquisition, improve technique, and enhance mental aspects of athletic performance.

Personal Development Applications

[0140] The TwinsDo system supports personal growth and development: [0141] Habit Formation: Users can observe their avatar successfully implementing desired habits and routines, strengthening neural pathways associated with these behaviors; [0142] Mindfulness and Stress Management: The system demonstrates meditation techniques, breathing exercises, and stress reduction strategies through the user's avatar; [0143] Time Management and Productivity: Users can learn organizational systems and productivity techniques by watching their avatar implement these strategies effectively; and [0144] Creative Skills: The system supports development of artistic, musical, and other creative abilities through demonstration and guided practice.

[0145] The personal development applications leverage the stimulation of neural mirroring mechanisms to help users internalize positive behaviors and mindsets, accelerating personal growth and transformation.

Content Management and Delivery

[0146] The TwinsDo system organizes educational and therapeutic content in a structured manner: [0147] Subject Organization: Content is categorized by subject area, including languages, academic disciplines, professional skills, personal development, and therapeutic protocols; [0148] Proficiency Levels: Within each subject, content is organized by proficiency level (beginner, intermediate, advanced) or therapeutic stage; [0149] Learning Modules: Each level contains multiple learning modules focused on specific skills, knowledge areas, or therapeutic objectives; and [0150] Lesson Structure: Modules are divided into lessons with clear learning objectives and assessment components.

[0151] Content is delivered through the user's personalized avatar, with the avatar appearing to teach the material directly to the user or demonstrate the skills being taught. This creates the impression that the user is watching themselves demonstrate mastery of the subject matter or skill, stimulating neural mirroring mechanisms that facilitate learning and skill development.

[0152] For users with special needs or preferences, content delivery includes additional considerations: [0153] Clear, concise language with minimal ambiguity; [0154] Visual supports to complement verbal information; [0155] Structured progression with predictable patterns; [0156]

Frequent positive reinforcement; [0157] Sensory considerations (adjustable volume, visual intensity, etc.); and [0158] Options to repeat or slow down content as needed

B2B Integration Via B2C Model

[0159] A unique aspect of the TwinsDo system is its approach to B2B integration through a B2C model: [0160] Content Exchange and Distribution Framework: Educational institutions, therapy providers, and content creators can develop and upload content to the TwinsDo marketplace; [0161] User Access: Individual users can purchase or subscribe to institution-provided content, accessing it through their personalized avatars; [0162] Credentialing System: Institutions can offer certificates or credentials through the platform, verifying user achievement and competency; [0163] Progress Sharing: Users can share their learning progress and therapeutic outcomes with affiliated institutions, with appropriate privacy controls; and [0164] Virtual Classrooms: Teachers and therapists can create virtual classrooms or therapy groups where their students or clients access specific content using their personalized avatars.

[0165] This approach allows educational institutions and therapy providers to leverage the neural mirroring stimulation capabilities of the TwinsDo system without developing their own technology infrastructure. Instead, they can focus on creating high-quality content that takes advantage of the system's unique capabilities.

[0166] For specialized providers, the marketplace offers opportunities to develop and distribute specialized protocols, reaching clients who may not have access to in-person services due to geographic, financial, or other constraints.

[0167] The TwinsDo system implements a multi-tiered access protocol with institutional distribution capabilities: [0168] Multi-Avatar Licensing: Educational institutions, healthcare providers, and corporate entities can purchase enterprise licenses that enable the creation and management of multiple avatars under a single account; [0169] Distributed Access: Institutional accounts can distribute access to their licensed avatars among students, patients, or employees who might not otherwise have the financial means to access the platform individually; [0170] Flexible Deployment: Organizations can customize the distribution and management of their licensed avatars by configuring usage parameters, access levels to content, and appropriate monitoring features. The TwinsDo platform provides enterprise-grade licensing models designed for institutions, allowing multiple user avatars to be created and managed under a single account. This approach supports large-scale deployment of neural-mirroring-based educational or therapeutic interventions while ensuring centralized control over costs, compliance and reporting; [0171] This model is especially suitable for schools, clinics, universities, corporations, and government programs that aim to reach broad populations. Institutional licenses include: [0172] a. Bulk avatar creation quotas; [0173] b. Administrative dashboards for usage and outcome tracking; [0174] c. Content access configuration by user group; [0175] d. Privacy controls and audit logs; and [0176] e. Analytics tools for curriculum or therapy plan optimization. [0177] 4. Distributed access protocols that extend educational content availability across socioeconomic barriers through institutional licensing frameworks; and [0178] 5. Usage Analytics: Institutional accounts receive detailed analytics on how their distributed avatars are being used, allowing for optimization of resource allocation and identification of high-impact content.

Marketplace for Institutional and Developer Content

[0179] In addition to institutional licensing, the TwinsDo platform includes a public-facing content marketplace where third-party developers, certified educators, therapists, and institutions can upload and monetize their own content. This marketplace supports both free and paid offerings, curated by subject, audience, and pedagogical objective. Contributors can: [0180] Create avatar-compatible educational modules, therapeutic programs, or coaching sequences; [0181] Link modules to credentialing or certification systems; [0182] Bundle content with usage-based or subscription-based access models; and [0183] Track performance and feedback metrics for their content within the platform.

[0184] This architecture enables a distributed business model where independent experts can build revenue-generating experiences using TwinsDo's core avatar-based neurostimulation engine.

Adaptive Learning System

[0185] The TwinsDo system includes an adaptive learning component that: [0186] Tracks User Progress: Monitors performance on exercises, assessments, and activities; [0187] Identifies Strengths and Weaknesses: Analyzes patterns in user performance to identify areas needing additional focus or areas of particular strength; [0188] Adjusts Content Difficulty: Dynamically modifies the difficulty level of presented material based on user performance and confidence; [0189] Personalizes Learning Path: Creates customized learning sequences based on individual needs, goals, learning style and pace; and [0190] Provides Targeted Feedback: Offers specific guidance on areas for improvement, delivered through the user's avatar to enhance receptivity. [0191] For users with special needs, the adaptive system includes additional parameters: [0192] Sensory preferences and sensitivities; [0193] Special interests that can be leveraged for engagement; [0194] Communication style and comprehension level; [0195] Attention span and optimal session duration; and [0196] Specific skill development goals.

[0197] The adaptive learning system ensures that users are consistently challenged at an appropriate level, maintaining engagement while maximizing learning outcomes and benefits.

Generative AI Integration

[0198] The TwinsDo system incorporates generative artificial intelligence capabilities to enhance personalization and effectiveness: [0199] Content Generation: AI models create customized educational materials based on user performance data, learning preferences, and goals; [0200] Dynamic Adaptation: The system continuously adjusts content complexity, presentation style, and pacing based on real-time user engagement metrics; [0201] Personalized Examples: The AI generates novel examples, explanations, and practice materials optimized for the individual user's learning profile; and [0202] Avatar Integration: All generated content is presented through the user's personalized avatar, maintaining the neural mirroring activation that is central to the system's effectiveness.

[0203] The generative AI components work in conjunction with the neural mirroring mechanisms to create a highly personalized learning experience that adapts to the user's needs while maintaining the neurological benefits of self-observation.

User Interface and Experience

[0204] The TwinsDo user interface is designed to be intuitive, accessible, and engaging: [0205] Dashboard: A personalized dashboard displays the user's avatar, progress metrics, recommended content, and quick access to favorite or recent activities; [0206] Content Browser: Users can explore available content by subject, level, or recommendation, with preview capabilities; [0207] Avatar Studio: A dedicated interface for creating, customizing, and updating the user's avatar; [0208] Learning/Therapy Space: The main interaction area where content is delivered through the user's avatar, with playback controls and interactive elements; [0209] Progress Tracker: Visual representations of learning progress, skill development, and achievement milestones; and [0210] Settings and Preferences: Controls for customizing the learning experience, including visual and auditory preferences, session duration, and notification settings.

[0211] For users with special needs, the interface includes accessibility features such as: [0212] Simplified layout options; [0213] High-contrast mode; [0214] Text-to-speech for navigation; [0215] Visual schedules and timers; [0216] Reduced animation options; and [0217] Customizable sensory settings.

[0218] The user experience is designed to minimize cognitive load while maximizing engagement with the content being presented through the user's avatar.

Data Collection and Privacy

[0219] The TwinsDo system collects various types of data to facilitate its functionality: [0220] User Profile Data: Basic demographic information, learning goals, and preferences; [0221] Biometric

Data: Images and voice recordings used to create the personalized avatar and voice model; [0222] Performance Data: User interactions, assessment results, and progress metrics; and [0223] Usage Patterns: Session frequency, duration, and content preferences.

[0224] To protect user privacy and comply with relevant regulations (such as GDPR, HIPAA, and COPPA), the system implements: [0225] Secure Data Storage: All data is encrypted at rest and in transit; [0226] Explicit Consent: Clear, age-appropriate consent processes for data collection and use; [0227] Data Minimization: Only collecting data necessary for system functionality; [0228] Access Controls: Strict limitations on who can access user data, with particular protections for children and vulnerable populations; [0229] Anonymization: Where possible, data used for system improvement is anonymized or pseudonymized; and [0230] Deletion Capabilities: Users can request deletion of their data, including biometric information used for avatar creation.

[0231] For therapeutic applications, additional safeguards ensure that sensitive health information is protected according to relevant healthcare privacy standards.

Implementation Technologies

[0232] The TwinsDo system utilizes a range of modern technologies to achieve its functionality:

[0233] Web Development: Frontend frameworks (React, Angular, or Vue.js) and backend technologies (Node.js, Python with Django/Flask, or Java with Spring) for the web application;

[0234] Artificial Intelligence and Machine Learning: TensorFlow, PyTorch, and specialized AI services for avatar generation, voice cloning, and adaptive learning algorithms; [0235] Computer

Vision: OpenCV and similar libraries for processing user images and creating 3D models; [0236]

Speech Processing: WaveNet-style deep learning models for voice analysis and synthesis; [0237]

3D Modeling and Animation: WebGL, Three.js, and related technologies for rendering and

animating avatars in the browser; [0238] Cloud Infrastructure: Scalable cloud services for processing, storage, and content delivery, with particular attention to handling the computational

demands of AI components; and [0239] Databases: A combination of relational databases for structured data and NoSQL solutions for flexible content storage.

[0240] The system architecture is designed to be modular and scalable, allowing for continuous improvement and the addition of new features as technology evolves.

Differentiation from Existing Technologies

[0241] The TwinsDo system is fundamentally different from existing educational and therapeutic technologies in several key ways: [0242] Personalized Avatar Focus: While platforms like Bitmoji, Memoji, or various avatar creators allow users to create digital representations of themselves, none systematically use these avatars as the primary delivery mechanism for educational or therapeutic content specifically to stimulate neural mirroring mechanisms; [0243] Neural Mirroring

Stimulation: Unlike general educational platforms (Coursera, Khan Academy, Duolingo) that focus on content delivery without consideration of neurological processes, TwinsDo is specifically designed to stimulate the activation of mirror neuron systems through self-observation; [0244]

Voice Cloning Integration: While some technologies offer voice synthesis or cloning (like ElevenLabs or Descript), none integrate this capability with personalized avatars for educational or therapeutic purposes; [0245] Cross-Domain Application: Unlike specialized platforms that focus on single domains (language learning, professional training, or therapy), TwinsDo provides a unified

approach to stimulating neural mirroring across multiple domains; [0246] Therapeutic Specificity: While some therapeutic technologies use avatars or virtual environments, none specifically leverage the user's own likeness to address mirror neuron deficits in conditions like ASD; [0247]

Athletic Performance Enhancement: Unlike conventional sports training technologies that focus on metrics and analysis, TwinsDo uniquely leverages neural mirroring mechanisms to accelerate skill acquisition and technique improvement; and [0248] Generative AI Integration: The system's incorporation of generative AI for content creation while maintaining the neural mirroring benefits represents a novel combination not found in existing educational or therapeutic technologies.

[0249] These differentiating factors create a unique and novel approach that has not been

implemented in any existing technology, providing substantial advantages in learning efficiency, engagement, and therapeutic efficacy.

Research and Validation

[0250] The TwinsDo system's effectiveness is supported by ongoing research and validation:

[0251] Comparative Studies: Controlled studies comparing learning outcomes between TwinsDo users and those using traditional methods or other educational technologies; [0252] Clinical Trials: For therapeutic applications, randomized clinical trials to assess efficacy compared to standard interventions; [0253] Neuroimaging Research: Studies using EEG, fMRI, or other neuroimaging techniques to directly observe neural activation patterns during use of the system; [0254] Longitudinal Analysis: Tracking user progress and outcomes over extended periods to assess long-term benefits and retention; and [0255] Qualitative Research: User interviews, focus groups, and observational studies to understand the subjective experience and identify areas for improvement. [0256] Research findings are continuously incorporated into the system design, creating an evidence-based approach to refinement and expansion of capabilities.

[0257] Preliminary pilot testing and controlled user observations indicate significantly higher engagement, retention, and user satisfaction when using the TwinsDo platform compared to conventional e-learning or therapeutic tools. These effects are especially pronounced among neurodivergent users, including individuals with Autism Spectrum Disorder. This early evidence supports the platform's core hypothesis that self-representation through realistic avatars and voice can facilitate measurable improvements in cognitive, behavioral, and emotional outcomes.

Future Developments

[0258] The TwinsDo system is designed with several planned future developments: [0259] Expanded Language Support: Adding additional languages beyond the initial English, Spanish, and Portuguese offerings; [0260] Enhanced Avatar Realism: Incorporating advances in computer graphics and AI to create increasingly realistic and expressive avatars; [0261] Voice Emotion Modeling: Enhancing the voice cloning technology to better capture and reproduce emotional nuances in speech; [0262] Mobile Applications: Developing dedicated mobile apps to complement the web-based platform; [0263] Virtual Reality Integration: Creating immersive VR experiences that further enhance the neural mirroring effect; [0264] Expanded Therapeutic Applications: Developing protocols for additional conditions beyond those initially supported; [0265] API Development: Creating application programming interfaces to allow third-party developers to build upon the platform's capabilities; [0266] Expanded Inclusive Access Models: Developing additional frameworks for democratizing access to the platform, including subsidized access programs for underserved communities, partnerships with public institutions, and sliding-scale pricing models based on regional economic factors; and [0267] Institutional Collaboration Tools: Creating specialized tools for educational and therapeutic institutions to collaborate on content development and share outcomes data while maintaining appropriate privacy and intellectual property protections.

[0268] These future developments will maintain the core focus on stimulating neural mirroring through personalized avatars while expanding the system's reach and capabilities.

[0269] The system implements biometric security protocols to ensure that each avatar is uniquely bound to its rightful user, preventing unauthorized reproduction of personal likeness. These protocols include cryptographically binding biometric identifiers to each avatar, implementing watermarking technology to prevent unauthorized use, providing secure storage with multi-factor authentication, and maintaining continuous verification protocols.

Claims

1. A system for stimulating the activation of neural mirroring mechanisms, comprising: An avatar creation module configured to generate a hyper-realistic or stylized digital avatar resembling a user

based on digital representations of the user's appearance; A voice cloning module configured to create a digital voice model that simulates or approximates the user's voice based on voice samples provided by the user; c. a content management module configured to organize educational or skill development content; An avatar animation module configured to animate the digital avatar to present the educational or skill development content using the digital voice model; A user interface configured to display the animated digital avatar to the user; and A progress tracking module configured to monitor and analyze the user's interaction with the system and learning outcomes.

2. The system of claim 1, wherein the avatar creation module is configured to: Process multiple images or video frames of the user to create a three-dimensional model of the user's face; Identify and map key facial features for animation; and Generate a digital avatar that maintains the user's distinctive facial characteristics while being suitable for animation.
3. The system of claim 1, wherein the voice cloning module is configured to: Analyze voice samples to identify the user's unique voice characteristics, including pitch, tone, rhythm, and speech patterns; Train a deep learning model to synthesize speech that mimics the user's voice; and Convert text from educational or skill development content into speech using the trained model.
4. The system of claim 1, wherein the progress tracking module is configured to: Analyze the user's performance on educational or skill development activities; Identify areas of strength and weakness; and Dynamically adjust content difficulty and presentation based on the user's progress and needs.
5. The system of claim 1, wherein the content management module is configured to organize content for multiple domains including: Academic education; Language learning; Professional skills development; Personal development; and Therapeutic applications.
6. The system of claim 1, further comprising a therapeutic protocol module configured to: Provide a progressive curriculum of skill development; Present training exercises through the user's avatar; Model appropriate interactions and behaviors; and Adjust content based on the user's specific needs and progress.
7. The system of claim 1, further comprising a content exchange and distribution framework configured to: Allow educational institutions, therapy providers, and human professionals to create, upload, and monetize specialized content; Enable users to access institution-provided and professional-created content through their personalized avatars; Facilitate sharing of progress data between users and affiliated institutions or professionals with appropriate privacy controls; Extend the reach of human professionals via avatar-driven asynchronous content delivery, preserving pedagogical integrity while increasing availability; Provide tools for professionals to extend their reach and effectiveness while maintaining their essential role in the educational or therapeutic process; and Provide institutional and enterprise licensing models that enable the creation and management of multiple avatars under single accounts, facilitating distributed access protocols that extend educational content availability across socioeconomic barriers through institutional licensing frameworks.
8. The system of claim 1, wherein the avatar animation module is configured to: Synchronize the avatar's lip movements with the synthesized speech; Generate appropriate facial expressions based on the emotional content of the material; and Incorporate natural gestures and body language to enhance engagement and comprehension.
9. The system of claim 1, further comprising an adaptive content generation module configured to: Utilize generative artificial intelligence models to create personalized educational content based on the user's learning patterns, preferences, and performance data; Dynamically adjust content complexity, presentation style, and pacing according to real-time user engagement metrics; Generate novel examples, explanations, and practice materials optimized for the individual user's learning profile; and Present all generated content through the user's personalized avatar to maintain neural mirroring activation.
10. The system of claim 1, further comprising a biometric security module configured to:

Cryptographically bind biometric identifiers to user accounts; Embed watermarking technology within avatar rendering to prevent unauthorized duplication; Securely store biometric templates with multi-factor authentication requirements for access; and Implement continuous verification protocols that monitor for unauthorized avatar usage or reproduction attempts.

11. A method for enhancing learning and skill development through stimulation of neural mirroring mechanisms, comprising: Generating a hyper-realistic or stylized digital avatar resembling a user based on digital representations based on the user's appearance; Creating a digital voice model that simulates or approximates the user's voice based on voice samples provided by the user; Organizing educational or skill development content appropriate for the user's needs and goals; Animating the digital avatar to present the educational or skill development content using the digital voice model; Displaying the animated digital avatar to the user through a user interface; and Tracking and analyzing the user's interaction with the system and learning outcomes.

12. The method of claim 11, wherein generating the digital avatar comprises: Processing multiple images or video frames of the user to create a three-dimensional model of the user's face; Identifying and mapping key facial features for animation; and Generating a digital avatar that maintains the user's distinctive facial characteristics while being suitable for animation.

13. The method of claim 11, wherein creating the digital voice model comprises: Analyzing voice samples to identify the user's unique voice characteristics including pitch, tone, rhythm, and speech patterns; Training a deep learning model to synthesize speech that mimics the user's voice; and Converting text from educational or skill development content into speech using the trained model.

14. The method of claim 11, further comprising adaptively adjusting the educational or skill development content by: Analyzing the user's performance on educational or skill development activities; Identifying areas of strength and weakness; and Dynamically adjusting content difficulty and presentation based on the user's progress and needs.

15. The method of claim 11, further comprising organizing content for multiple domains including: Academic education; Language learning; Professional skills development; Personal development; and Therapeutic applications.

16. The method of claim 11, further comprising implementing therapeutic protocols for users with special needs by: Providing a progressive curriculum of skill development; Presenting training exercises through the user's avatar; Modeling appropriate interactions and behaviors; and Adjusting content based on the user's specific needs and progress.

17. The method of claim 11, further comprising facilitating a content exchange and distribution framework by: Allowing educational institutions, therapy providers, and human professionals to create, upload, and monetize specialized content; Enabling users to access institution-provided and professional-created content through their personalized avatars; Facilitating sharing of progress data between users and affiliated institutions or professionals with appropriate privacy controls; Extending the reach of human professionals via avatar-driven asynchronous content delivery, preserving pedagogical integrity while increasing availability; Providing tools for professionals to extend their reach and effectiveness while maintaining their essential role in the educational or therapeutic process; and Providing institutional and enterprise licensing models that enable the creation and management of multiple avatars under single accounts, facilitating distributed access protocols that extend educational content availability across socioeconomic barriers through institutional licensing frameworks.

18. The method of claim 11, further comprising animating the digital avatar by: Synchronizing the avatar's lip movements with the synthesized speech; Generating appropriate facial expressions based on the emotional content of the material; and Incorporating natural gestures and body language to enhance engagement and comprehension.

19. The method of claim 11, further comprising generating adaptive content by: Utilizing generative artificial intelligence models to create personalized educational content based on the user's learning patterns, preferences, and performance data; Dynamically adjusting content

complexity, presentation style, and pacing according to real-time user engagement metrics; Generating novel examples, explanations, and practice materials optimized for the individual user's learning profile; and Presenting all generated content through the user's personalized avatar to maintain neural mirroring activation.

20. The method of claim 11, further comprising implementing biometric security by:

Cryptographically binding biometric identifiers to user accounts; Embedding watermarking technology within avatar rendering to prevent unauthorized duplication; Securely storing biometric templates with multi-factor authentication requirements for access; and Implementing continuous verification protocols that monitor for unauthorized avatar usage or reproduction attempts.

21. A system for enhancing athletic performance through stimulation of neural mirroring mechanisms, comprising: An avatar creation module configured to generate a hyper-realistic or stylized digital avatar resembling an athlete based on digital representations based on the athlete's appearance; A voice cloning module configured to create a digital voice model that simulates or approximates the athlete's voice based on voice samples provided by the athlete; A sports training content module configured to organize training materials for specific sports techniques, strategies, mental preparation, and performance optimization; An avatar animation module configured to animate the digital avatar to demonstrate proper athletic techniques and performance strategies using the digital voice model; and A performance tracking module configured to monitor the athlete's skill acquisition, technique improvement, and performance metrics in real-world athletic contexts.
